

Vizualization

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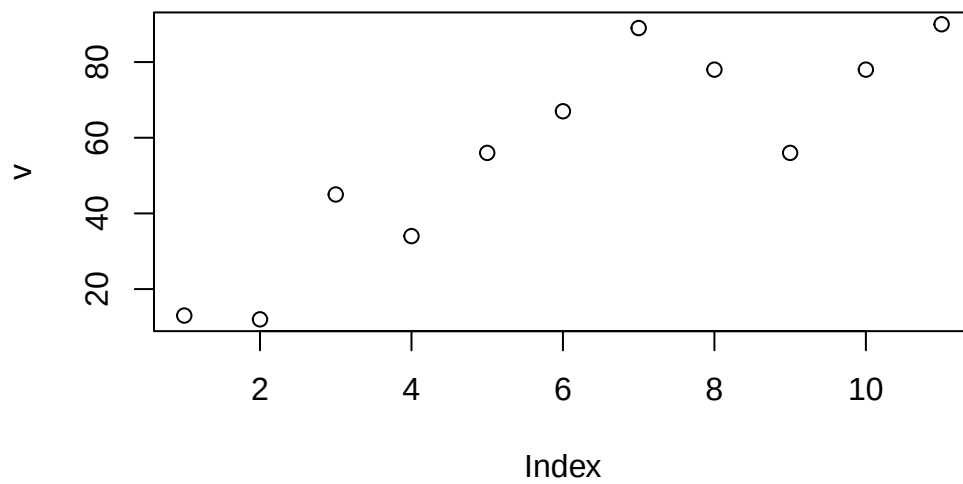
Table of contents

Line graph	1
Bar Graph	5
ggplot	6

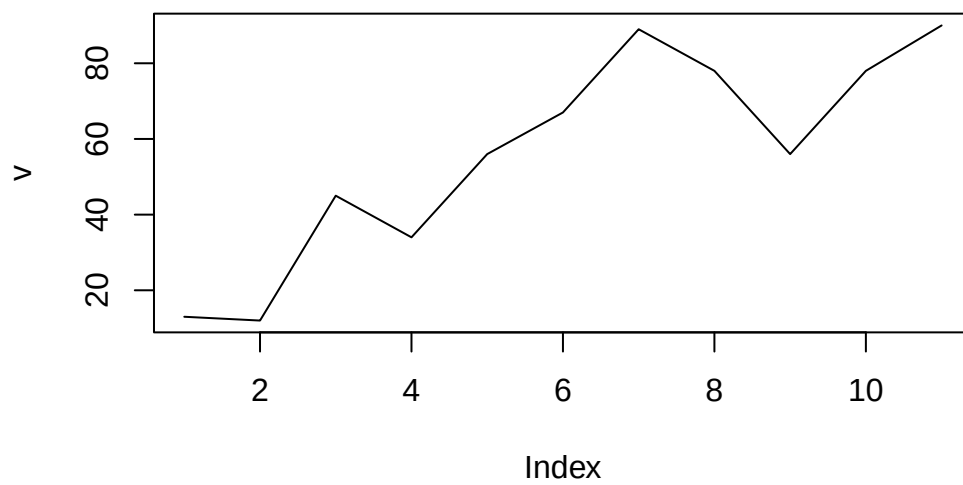
Line graph

Basics

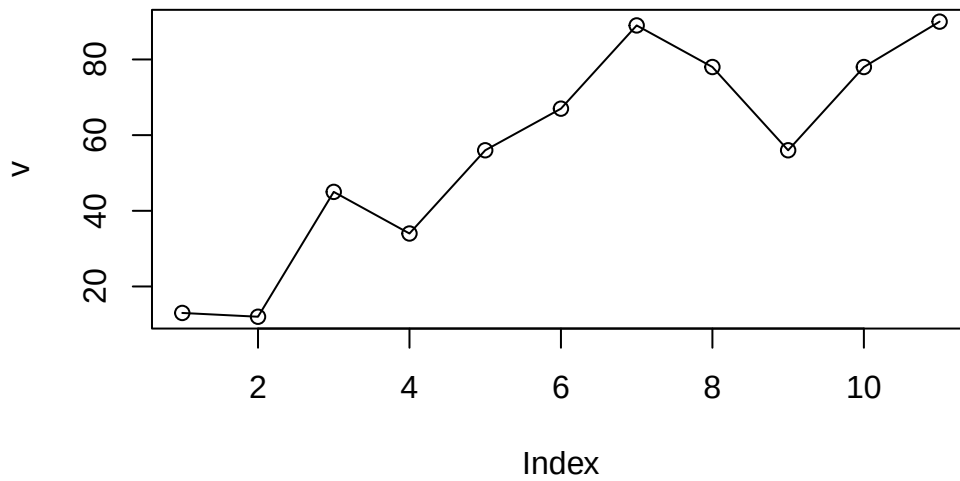
```
v <- c(13,12,45,34,56,67,89,78,56,78,90)
#this is just create point
plot(v,type = "p")
```



```
#this is just create line  
plot(v,type = "l")
```



```
#this is create both line and point  
plot(v,type = "o")
```



With the headline

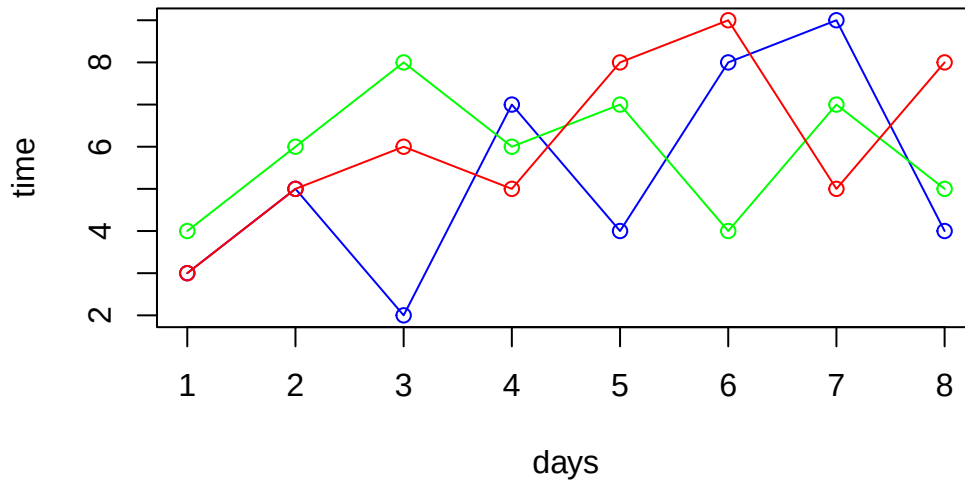
```
days = c(10,20,30,40,50,60,70,80,90)  
income <- c(1000,3000,4500,6700,7800,8900,9900,11000,12000)  
plot(days,income,type = "o",main = "Salary",xlab = "days",ylab = "income", col="green")
```



Multiple Graph

```
f <- c(3,5,2,7,4,8,9,4)
g <- c(4,6,8,6,7,4,7,5)
j <- c(3,5,6,5,8,9,5,8)
plot(f, type="o",main="multiple graph",xlab="days",ylab="time",col="blue")
lines(g, type="o",col="green")
lines(j, type="o",col="red")
```

multiple graph



Bar Graph

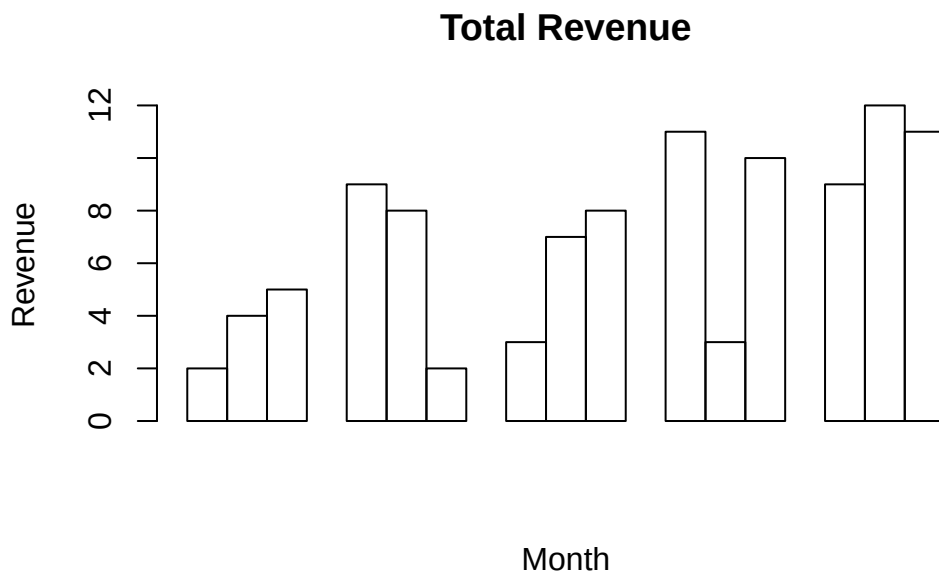
```
#Create the matrix of the values
values <- matrix(c(2, 9, 3, 11, 9, 4, 8, 7, 3, 12, 5, 2, 8, 10, 11),nrow=3, ncol=5, byrow=TRUE)
#Create the bar chart
barplot(values, main="Total Revenue", name.arg=months, xlab="Month",ylab="Revenue", col = c("red", "green", "blue"))
```

Warning in plot.window(xlim, ylim, log = log, ...): "name.arg" is not a graphical parameter

Warning in rect(y1, x1, y2, x2, ...): supplied colour is neither numeric nor character

Warning in title(main = main, sub = sub, xlab = xlab, ylab = ylab, ...): "name.arg" is not a graphical parameter

Warning in axis(if (horiz) 1 else 2, cex.axis = cex.axis, ...): "name.arg" is not a graphical parameter



ggplot

```
data <- data.frame(  
  x = c(1, 2, 3, 4, 5),  
  y = c(3, 5, 2, 8, 7)  
)
```

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
ggplot(data,aes(x,y))+  
  geom_line(color= "blue")+  
  geom_point(size=3)+  
  labs(title="Practice of ggplot", subtitle="add any subtitle hare",caption="add any caption")  
  theme_classic()
```

